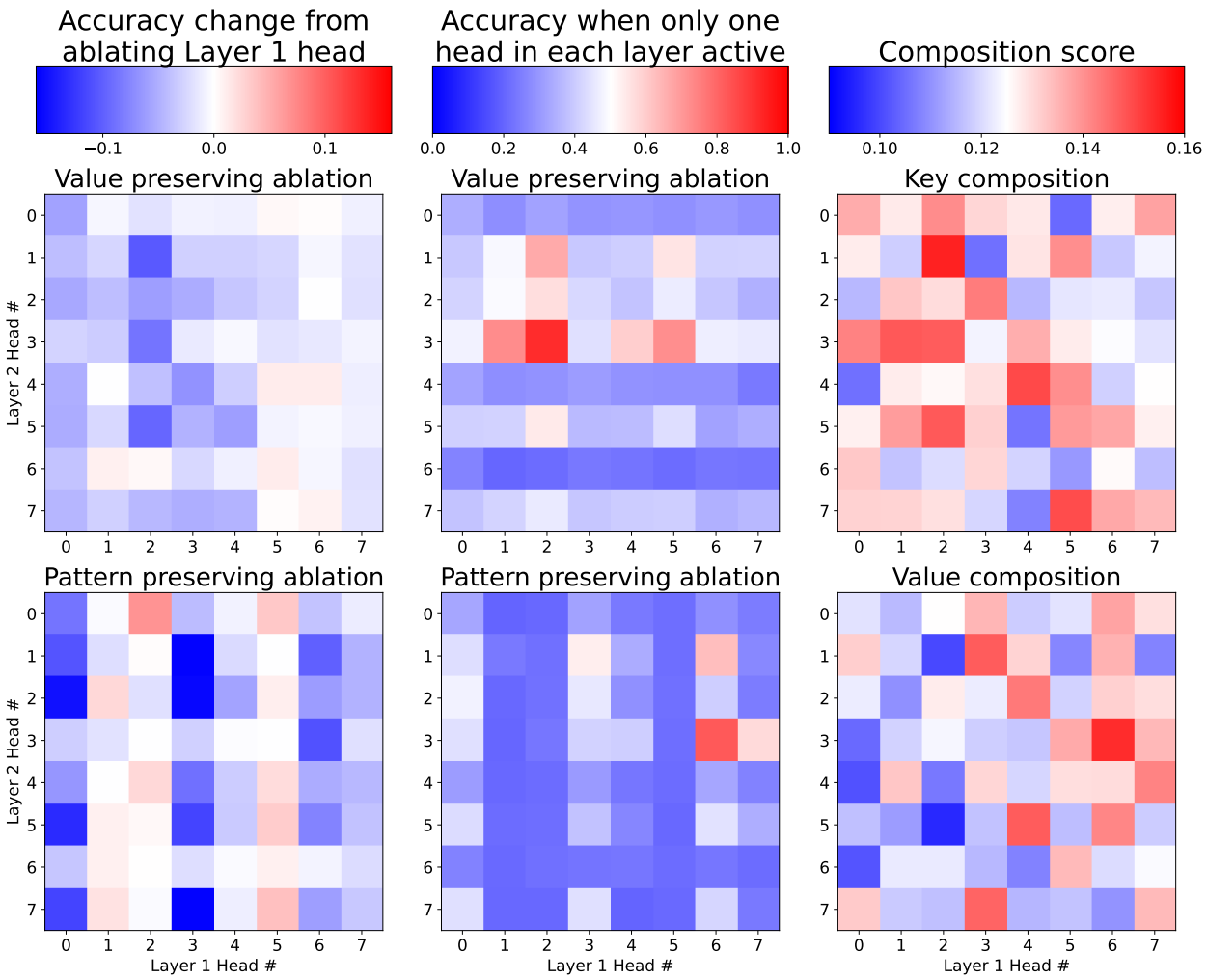

Figure 11. Layer 1 → Layer 2 head connectivity analyses (three different methods as the three columns). **Leftmost column:** Effect of ablating a single Layer 1 head. Each column corresponds to the Layer 2 head that is kept active. In this plot, we plot the deviation compared to if the given Layer 1 head was not ablated, to emphasize positive/negative differences (here darker blue would mean more influential). Left plot preserves values, so results will be most affected by Layer 1 heads that contribute to the match subcircuit (previous token heads). From this, we see that Layer 1 Head 2 is an important previous token head for many of the Layer 2 heads. The right plot preserves patterns, so results will be least affected by Layer 1 heads that only contribute to the match subcircuit (previous token heads). In fact, we see that when ablating heads 1, 2, and 5, performance improves, indicating that previous token heads might be interfering with the network in some cases due to (suboptimal) V-composition. **Middle column:** Performance of a single pair of heads (one from Layer 1, and one from Layer 2) when preserving Layer 2 values (left) or patterns (right). In this case, we plot the raw accuracy to show how strong each path through the network is. The color scheme highlights points that go above simple context copying strategies (which has an accuracy of 50%) in red. We see that Layer 1 Head 2 → Layer 2 Head 3 is the strongest path (when preserving values), leading to a performance of 90%. The left plot indicates that Layer 1 heads 1, 2, and 5 play a strong role in computing patterns. The right plot indicates that Layer 1 Head 6 has the most important V-composition to Layer 2 heads. **Rightmost column:** For comparison, we include the composition score (Nanda, 2022; Elhage et al., 2021) of Layer 1 head output projections to key and value input matrices in Layer 2. We again use a bwr colorscheme to highlight pairs with higher than average composition scores. While the main forms of composition are still visible (K-composition from Layer 1 Head 2 to Layer 2 Head 1/3, and V-composition from Layer 1 Head 6 to Layer 2 Head 3), we note that the results are overall way noisier. This suggests that the activation modification approach used in the middle column, made easy to implement by our artificial optogenetics framework, may provide clearer signals than weight-based approaches on circuit wiring.

F. Proof of saddle point in toy model

We include a brief proof that $\mathbf{a} = \mathbf{b} = \mathbf{c} = \mathbf{0}$ is a saddle point for the toy model, leading to the long loss plateau before the phase change.

Recall, the loss being minimized is:

$$\mathcal{L}(\mathbf{a}, \mathbf{b}, \mathbf{c}) = \frac{1}{2} \sum_{i,j,k} \left(a_i^{true} b_j^{true} c_k^{true} - a_i b_j c_k \right)^2,$$

and we assume $\mathbf{a}^{true}, \mathbf{b}^{true}, \mathbf{c}^{true} \neq \mathbf{0}$.

We then have:

$$\frac{\partial \mathcal{L}(\mathbf{a}, \mathbf{b}, \mathbf{c})}{\partial a_i} = -\sum_{j,k} \left(a_i^{true} b_j^{true} c_k^{true} - a_i b_j c_k \right) b_j c_k = 0,$$
$$\frac{\partial \mathcal{L}(\mathbf{a}, \mathbf{b}, \mathbf{c})}{\partial b_j} = -\sum_{i,k} \left(a_i^{true} b_j^{true} c_k^{true} - a_i b_j c_k \right) a_i c_k = 0,$$
$$\frac{\partial \mathcal{L}(\mathbf{a}, \mathbf{b}, \mathbf{c})}{\partial c_k} = -\sum_{i,j} \left(a_i^{true} b_j^{true} c_k^{true} - a_i b_j c_k \right) a_i b_j = 0,$$

if $\mathbf{a} = \mathbf{b} = \mathbf{c} = \mathbf{0}$. Moving forward, we will abbreviate derivations using the symmetry between a, b, c.

The value of the loss function at this point is:

$$\mathcal{L}(\mathbf{0}) = \frac{1}{2} \sum_{i,j,k} \left(a_i^{true} b_j^{true} c_k^{true} \right)^2$$

While we know this point is neither a global maximum (∞) nor a global minimum (0) of the loss function, we now need to show it's a saddle point (and not a local minimum/maximum).

F.1. Hessian test inconclusive for 3 or more interacting variables

A first attempt to show it's a saddle point (and not a local minimum/maximum) would be to use the determinant of the Hessian. We note that such a proof does work in the two vector case (the determinant of the Hessian is negative in that case), but for the 3-variable case, the determinant of the Hessian is also 0. We show this below:

The Hessian can be seen as a 3×3 block matrix:

$$\mathcal{H} = \begin{pmatrix} H_{aa} & H_{ab} & H_{ac} \\ H_{ba} & H_{bb} & H_{bc} \\ H_{ca} & H_{cb} & H_{cc} \end{pmatrix},$$

and we can work out the individual terms below:

$$\frac{\partial^2 \mathcal{L}(a, b, c)}{\partial a_i^2} = \sum_{j,k} (b_j c_k)^2 = 0$$
$$\frac{\partial^2 \mathcal{L}(a, b, c)}{\partial a_i \partial a_{i'}} = 0, i \neq i'$$

so $H_{aa} = H_{bb} = H_{cc} = \mathbf{0}$. For the cross terms, we have:

$$\frac{\partial^2 \mathcal{L}(a, b, c)}{\partial a_i \partial b_j} = \sum_k 2 a_i b_j c_k^2 - a_i^{true} b_j^{true} c_k^{true} c_k = 0$$

Here, the presence of c_k forces the these partials to 0, making the Hessian a matrix of all 0's. Note in the two variable case (if we remove $\mathbf{c}$), the Hessian would have positive off-diagonal elements and it becomes easy to show the determinant is negative.

F.2. Directions of positive and negative slope

Since the Hessian test doesn't work, we can instead attempt to identify two directions, one in which the function will increase in a small neighborhood and one in which it will decrease in a small neighborhood. These are easy to construct by picking i', j', k' s.t. $a_{i'}^{true} b_{i'}^{true} c_{i'}^{true} \neq 0$ (such indices are guaranteed to exist since $\mathbf{a}^{true}, \mathbf{b}^{true}, \mathbf{c}^{true} \neq \mathbf{0}$).

Specifically, consider the direction of $\mathbf{d}_- = \langle 0, ..., 0, a_{i'}^{true}, 0, ..., 0, b_{j'}^{true}, 0, ..., 0, c_{k'}^{true}, 0, ..., 0 \rangle$. We can see that a small $\varepsilon > 0$[22] perturbation along this direction will reduce the loss:

$$
\begin{aligned}
\mathcal{L}(\mathbf{0} + \varepsilon \mathbf{d}_-) &= \frac{1}{2} \sum_{i,j,k} \left(a_i^{true} b_j^{true} c_k^{true} - a_i b_j c_k \right)^2 \\
&= \frac{1}{2} \left(a_{i'}^{true} b_{j'}^{true} c_{k'}^{true} - \varepsilon^3 a_{i'}^{true} b_{j'}^{true} c_{k'}^{true} \right)^2 + \frac{1}{2} \sum_{i \neq i', j \neq j', k \neq k'} \left(a_i^{true} b_j^{true} c_k^{true} \right)^2 \\
&= \frac{1}{2} \left(a_{i'}^{true} b_{j'}^{true} c_{k'}^{true} \right)^2 (\varepsilon^6 - 2\varepsilon^3) + \frac{1}{2} \sum_{i,j,k} \left(a_i^{true} b_j^{true} c_k^{true} \right)^2 \\
\Rightarrow \lim_{\varepsilon \to 0} \mathcal{L}(\mathbf{0} + \varepsilon \mathbf{d}_-) - \mathcal{L}(\mathbf{0}) &= \frac{1}{2} \left(a_{i'}^{true} b_{j'}^{true} c_{k'}^{true} \right)^2 (\varepsilon^6 - 2\varepsilon^3) \approx -\varepsilon^3 \left(a_{i'}^{true} b_{j'}^{true} c_{k'}^{true} \right)^2 < 0
\end{aligned}
$$

Thus, a direction with a negative directional derivative exists.

Similarly, we consider $\mathbf{d}_+ = \langle 0, ..., 0, -a_{i'}^{true}, 0, ..., 0, b_{j'}^{true}, 0, ..., 0, c_{k'}^{true}, 0, ..., 0 \rangle$. We can see that a small $\varepsilon > 0$ perturbation along this direction will increase the loss:

$$
\begin{aligned}
\mathcal{L}(\mathbf{0} + \varepsilon \mathbf{d}_+) &= \frac{1}{2} \sum_{i,j,k} \left(a_i^{true} b_j^{true} c_k^{true} - a_i b_j c_k \right)^2 \\
&= \frac{1}{2} \left(a_{i'}^{true} b_{j'}^{true} c_{k'}^{true} + \varepsilon^3 a_{i'}^{true} b_{j'}^{true} c_{k'}^{true} \right)^2 + \frac{1}{2} \sum_{i \neq i', j \neq j', k \neq k'} \left(a_i^{true} b_j^{true} c_k^{true} \right)^2 \\
&= \frac{1}{2} \left(a_{i'}^{true} b_{j'}^{true} c_{k'}^{true} \right)^2 (\varepsilon^6 + 2\varepsilon^3) + \frac{1}{2} \sum_{i,j,k} \left(a_i^{true} b_j^{true} c_k^{true} \right)^2 \\
\Rightarrow \lim_{\varepsilon \to 0} \mathcal{L}(\mathbf{0} + \varepsilon \mathbf{d}_+) - \mathcal{L}(\mathbf{0}) &= \frac{1}{2} \left(a_{i'}^{true} b_{j'}^{true} c_{k'}^{true} \right)^2 (\varepsilon^6 + 2\varepsilon^3) \approx \varepsilon^3 \left(a_{i'}^{true} b_{j'}^{true} c_{k'}^{true} \right)^2 > 0
\end{aligned}
$$

Thus, we've identified two directions, one which increases the loss and one which decreases the loss, indicating that $\mathbf{a} = \mathbf{b} = \mathbf{c} = \mathbf{0}$ must be a saddle point. Note, if we find the directional derivative in either of these directions, it's still 0 (due to the factor of ε^3 not vanishing, even if we divide by ε to get the derivative)—we simply show the derivation to show the differing sign. We also note that this argument extends beyond 3 interacting vectors.

This argument can also give some intuition as to why the saddle point is "harder to escape" in the case of 3 interacting vectors (as evidenced by the longer loss plateau in the black curve versus the blue curve in Figure 5a). Here, we have a factor of ε^3 rather than ε^2 (in the two vector case). The intuition extends: if the number of components is increased further (beyond 3), the saddle point will get harder and harder to escape.

G. Assorted supplementary figures

While writing the main paper, we tried to maintain a coherent narrative that highlights the main results and intuitions we want readers to take away. For the in-the-trenches researcher, we provide some additional results that we thought were interesting.

[22] Note, the positivity assumption is not necessary. In fact, if we allow ε to take either sign, we can just use one direction to see the saddle point (due to the factor of ε^3). However, for the case of an even number of components (e.g., $N = 4$ instead of $N = 3$), we do need two separate directions (since ε^N will always be positive if N is even). To make this proof general, we thus provide the two directions and focus on positive ε.